Louai Aldalati

Software Developer

louai.workmail@gmail.com Turkey, Gaziantep louai-aldalati.github.io/louai_aldalati_portfolio

PROFILE SUMMARY

Software Engineering student specializing in developing integrated software solutions that merge the power of Backend (Django) with AI models (LLMs) and Computer Vision. Characterized by analytical thinking and fluency in three languages (Arabic, English, Turkish). Highly motivated to leverage modern technologies to innovate efficient systems and solve complex real-world problems.

SKILLS

Programming Languages
Python, Java, C++, C, JavaScript, Dart, SQL.

AI & Machine Learning
NLP, Computer Vision, RAG, Fine-Tuning, Transfer Learning, Deep Reinforcement Learning (DRL), Time-Series Analysis.

Mobile Development
Flutter, Cross-Platform Development, Google Maps API.

Web & Databases
Django, Django REST Framework, HTML, CSS, API Development, Relational Databases, PostgreSQL, Stored Procedures & Triggers, NoSQL & Firebase.

Models & Tools
Docker, Llama 3, MobileNetV2, XGBoost, Git, GitHub, Agile/Scrum.

Soft Skills
Analytical Thinking, Teamwork, Effective Communication, Continuous Learning.

EDUCATION

Bachelor's in Software Engineering 2023 – 2027
Manisa Celal Bayar University
GPA: 3.78 / 4.00

High School - Science Stream 2020 – 2023
Çamlık Anadolu Lisesi

TECHNICAL PROJECTS

AI Hotel Chatbot 09/2025 – 01/2026
Technologies: Llama 3 (8B Instruct), RAG, Fine-Tuning, NLP
- Developed an advanced intelligent assistant based on the open-source Llama 3 model to provide accurate and automated responses.
- Applied Retrieval-Augmented Generation (RAG) and Fine-Tuning to fully customize model responses based on specific hotel policies and data, enhancing customer service efficiency.
- Source Code: https://github.com/louai-aldalati/sunrise_hotel

Real-Time Emotion Recognition 09/2025 – 01/2026
Technologies: Computer Vision, MobileNetV2, Transfer Learning, Python
- Built a local computer vision system that analyzes live webcam feeds to classify 8 different human emotions with high accuracy.
- Utilized Transfer Learning with the MobileNetV2 architecture, fine-tuning final layers to accelerate performance and reduce resource consumption.
- Source Code: https://github.com/louai-aldalati/Emotion-Recognition-MobileNetV2

Intelligent Cloud HR (Candidate Filter) 02/2026 – 06/2026
Technologies: Python, Django, spaCy (NLP), Tesseract OCR, PostgreSQL, Docker, Railway PaaS
- Developed a cloud-based HR platform leveraging Natural Language Processing (NLP) and Optical Character Recognition (OCR) to automate CV data extraction and generate an objective "Match Score" against job criteria.
- Built the backend using Django and PostgreSQL, and deployed the system via Docker containers on the Railway PaaS to ensure security and scalability.

- Source Code: https://github.com/louai-aldalati/candidate_filter

Smart Resource Allocation System 02/2026 – 06/2026
Technologies: Deep Reinforcement Learning (DQN), Python, PyTorch, Gymnasium, Algorithm Design
- Engineered a simulated cloud environment to solve the dynamic 2D Knapsack problem, optimizing CPU and RAM allocation for incoming tasks.
- Trained a Deep Q-Network (DQN) agent with comprehensive hyperparameter optimization, outperforming traditional heuristic (Greedy) algorithms and achieving 90.3% of the theoretical maximum (Dynamic Programming).
- Source Code: https://github.com/selinkincal/smart-resource-allocation-rl

Weather Forecasting ML API 09/2025 – 01/2026
Technologies: Machine Learning, XGBoost, Time-Series Analysis, Django, DRF
- Designed and trained a machine learning model using XGBoost to analyze time-series data (temperature, humidity, wind direction) for future value prediction.
- Developed the backend using Django, providing the AI model as a service via Django REST API for seamless integration with web or mobile apps.
- Source Code: https://github.com/louai-aldalati/weather-predictor

Pawly: Pet Care Marketplace Mobile App 02/2026 – 06/2026
Technologies: Flutter, Dart, Firebase (Auth, Firestore, FCM), Google Maps API, NoSQL
- Developed a comprehensive cross-platform mobile marketplace connecting pet owners with service providers, featuring a dynamic multi-role architecture (Pet Owner, Service Provider, Hybrid) within a single application.
- Engineered a real-time backend utilizing Firebase Firestore, and seamlessly integrated advanced features including Google Maps API with Geocoding for location tracking, and Firebase Cloud Messaging (FCM) for instant push notifications.
- Source Code: https://github.com/selinkincal/pet_care_app

Kelime6: Language Learning Platform 02/2025 – 06/2025
Technologies: Django, Python, Relational Database, Git, Scrum
- Programmed an educational system based on the Spaced Repetition System (SRS) algorithm to schedule word appearances based on user performance.
- Developed modules for user authentication, analytical dashboards, and a dynamic testing system including an interactive Wordle game.
- Source Code: https://github.com/louai-aldalati/kelime6

TinyHouse: Rental Management System 02/2025 – 06/2025
Technologies: Django, SQL (Stored Procedures, Triggers), RBAC
- Developed a property management platform supporting advanced Role-Based Access Control (RBAC) for Admins, Hosts, and Guests.
- Engineered a normalized relational database utilizing Stored Procedures and Triggers to automate booking and payment updates.
- Source Code: https://github.com/louai-aldalati/tinyhouse

LANGUAGES

Arabic
Native

Turkish
Fluent

English
Advanced

TRAINING COURSES

Monitoring and Evaluation in NGOs 2023
Trainer Khalil M. Hasno
Covers key concepts in performance monitoring and evaluation (M&E) and provides practical insights into professional reporting processes.

Social Media Marketing Strategies 2020
Gaziantep Chamber of Commerce
Explores the core principles and strategies for digital marketing and effective brand positioning across social media platforms.

Basics of Social Media 2020
Alqadasi Business
Focuses on key e-commerce concepts and strategies for maximizing online profitability and digital presence.

Developing Entrepreneurship and Innovation 2019
INJAZ Lebanon
Addresses the fundamental traits of entrepreneurship and critical success factors for establishing and managing new projects.

Life Skills 2019
AVSI
Explores core topics regarding human nature, psychological structure, and personal development.

Electrical Sciences 2019
AVSI
Introduces fundamental electrical concepts with a focus on practical applications in residential electrical systems.

Technology and Information Capabilities 2017
Dot Lebanon
Covers essential computer usage concepts and proficiency in Microsoft Office applications.

HOBBIES

Chess — Enhancing strategic thinking and problem-solving skills. | **Programming** — Constantly building side projects and exploring new frameworks. | **Cycling** — Maintaining physical fitness and endurance. | **Sports** — General interest in fitness and athletic activities.

REFERENCES

Assoc. Prof. Dr. Fatih Yücalar, *Head of Computer Software Department,*
Hasan Ferdi Turgutlu Faculty of Technology, Manisa Celal Bayar University
fatih.yucalar@cbu.edu.tr

Asst. Prof. Dr. İrfan Aygün, *Computer Hardware Department,*
Hasan Ferdi Turgutlu Faculty of Technology, Manisa Celal Bayar University
irfan.aygun@cbu.edu.tr